HONG KONG BAPTIST UNIVERSITY
COURSE OUTLINE

1. COURSE TITLE

Computer Vision

2. COURSE CODE

COMP7055

3. NO. OF UNITS

3 Units

4. OFFERING DEPARTMENT

Master of Science in Data Analytics and Artificial Intelligence

5. PREREQUISITES

Nil

6. MEDIUM OF INSTRUCTION

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7. AIMS & OBJECTIVES

To give students a comprehensive knowledge on computer vision, to discuss recent research advancements in selected computer vision topics, to design and develop a computer vision application prototype.

8. COURSE CONTENT

I. Introduction to Computer Vision
- Background, requirements and issues, human vision, applications

II. Image Formation: Geometry and Photometry
- Geometry, photometry (brightness and color), quantization, camera calibration

III. Image Segmentation and Feature Extraction: Conventional Methods
- Various methods of image segmentation, edge detection, object proposals

IV. Multi-view Geometry
- Shape from stereo and motion, feature matching

V. Object Detection: Conventional Methods
- HoG/SIFT features, Bayes classifiers, SVM classifiers

VI. Introduction to Neural Networks
- Key components and basic architecture of deep neural network, loss functions, backpropagation and SGD, Batch Normalization

VII. Convolution Neural Networks and Vision Transformers
- Famous CNN architectures, training strategies
Basic Architecture and Developments of Vision Transformers

VIII. Object Detection: Deep Learning Methods
- Background and principles of R-CNN, fast/faster R-CNN

IX. Image Segmentation: Deep Learning Methods
- Seminal architectures for image segmentation, e.g., auto-encoder networks, U-Net, Mask R-CNN

X. Temporal Processing and Recurrent Neural Network
- Background and principles of RNN, image captioning, action recognition, motion prediction applications

9. COURSE INTENDED LEARNING OUTCOMES (CILOs)

CILO	By the end of the course, students should be able to:
CILO 1	Explain basic theories and techniques in computer vision.
CILO 2	Identify various approaches of computer vision, and design the components of computer vision systems.
CILO 3	Describe and discuss the basic functions and methods for image processing.
CILO 4	Design and develop a computer vision application prototype.

10. TEACHING & LEARNING ACTIVITIES (TLAs)

CILO alignment	Type of TLA
1-3	Students will learn knowledge of computer vision through lectures and tutorials. In order to help students to have good understanding, laboratory sessions will be designed so that students could apply what they have learnt in lectures. Besides, written assignment(s), laboratory exercise(s) and final examination will be designed to evaluate the students' level of understanding.
2,4	Based on the theories of computer vision they have learnt, students are required to develop an application prototype using an API, such as

	OpenCV. Students are required to give a preliminary demonstration as well as a final formal presentation on their project. In both cases, instructor(s), teaching assistant and other students would ask questions related to their project. In this way, we could assess their understanding of the theories of computer vision and pattern recognition, as well as their proposed method.

11. ASSESSMENT METHODS (AMs)

Type of Assessment Methods	Weighting	CILOs to be addressed	Description of Assessment Tasks
Continuous Assessment	50 %	1-4	Continuous assessments, including written assignment, laboratory exercises and projects are designed to measure how well students have learned the fundamentals and major concepts of computer vision.
Examination	50 %	1-3	Final examination questions are designed to see how far students have achieved their course intended learning outcomes.

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